

CAN WE FEED LESS STARCH TO OUR COWS?

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The large increase in corn price in the last 12-15 months has increased the interest in lowering feed costs by substituting other feeds for some of the corn fed in dairy rations. Corn grain is typically used as the primary carbohydrate energy source in many dairy rations in the U.S. The starch content of corn grain is 70 – 75% on a dry matter basis. Thus, corn grain provides a concentrated starch source in rations. Even though the price of corn has increased, a recent analysis indicated that corn grain was still a bargain relative to the nutrients it provides in dairy cattle rations (St. Pierre, 2007).

It is important to realize that starch is not a required nutrient in ruminant rations for either the animal or the rumen microorganisms. The rumen microorganisms need a supply of rumen fermentable carbohydrates as an energy source for growth and microbial protein synthesis. A number of feed components can supply fermentable carbohydrates including starch, sugar, pectin and digestible fiber. Starch can be digested in the rumen as a fermentable carbohydrate source. The starch that reaches the intestine intact can also be utilized as a glucose source for milk production or energy.

STARCH CONTENT OF FEEDS

Table 1 contains the starch content of some common feeds used in dairy rations. These are “typical” values. Individual samples could vary based on source or processing.

Table 1. Starch Content of Common Feeds

Feed	Starch, % of DM
Corn grain	70 -75
Barley grain	50 – 56
Oat grain	40 – 44
Wheat grain	62 – 65
Hominy	50 – 55
Wheat midds	18 – 26
Wheat bran	22 – 26
Ear corn	55 – 62
Corn gluten feed	14 – 18
Corn gluten meal	15 – 18
Corn germ meal	15 – 18
Corn silage	20 – 45

RATION STARCH LEVELS

Dairy cattle rations currently fed usually contain between 20 and 30% starch on a dry matter basis. What about rations fed to high producing dairy herds? A survey of feeding programs in high producing herds was recently reported (Chase, 2006). This survey included 7 Jersey herds (> 17,000 lbs milk) and 25 Holstein herds (> 29,000 lbs. milk) from across the U.S. Ration starch levels ranged between 22 and 27% for the Jersey herds and 21 to 30% for the Holstein herds. The majority of these herds were in the East and Midwest regions of the U.S. Reported ration starch levels were 25 to 29% in 6 Wisconsin Holstein herds producing > 29,000 lbs. of milk per cow (Shaver and Kaiser, 2004). The starch content of these rations accounted for 53 to 74% of the total nonfiber carbohydrates (NFC). Hall and Van Horn (2001) surveyed U.S. dairy herd rations that were associated with high milk production and good health. Ration starch levels ranged from about 15 to 27%. The higher starch level rations tended to be in herds feeding higher levels of forage in the total ration. The average starch content of 56 total mixed ration samples analyzed in 2007 by the Marshfield Soil and Forage Analysis lab was 25.88%. The standard deviation of these samples was 4.19.

Ration starch levels ranged from 17 to 25% in 7 Holstein herds in Washington state producing >28,000 lbs. of milk (Johnson et.al., 2002). These herds feed less corn silage as a percent of the total forage than most of our Northeast herds. These herds also feed higher proportions of grass or alfalfa forages and use more dry hay than herds in the Northeast. Three example rations from Israel targeted for cows producing about 90 lbs. of milk were provided by Solomon (2007). These rations ranged in starch content from 21 to 25%.

STARCH DEGRADABILITY

It is not sufficient to look only at the ration starch level. The quantity of intake starch that is fermented and available in the rumen must also be considered. Even though this approach makes sense biologically, it is difficult to routinely implement in the field. The primary problem is that there is no accepted and commercially available analytical test to measure ruminal starch degradability (RDS). A variety of in vitro, in situ, gas production and enzyme methods have been used and reported in research literature. The diversity of approaches used in research makes it difficult to compare results between papers. The results obtained from most of these approaches have not been calibrated with animal performance data.

Lykos and Varga (1995) examined the relationship between particle size and ruminal starch degradability. Table 2 contains the results of this trial. Effective rumen degradability increases as corn particle size decreases. Steam-flaked corn does not fit the particle size pattern. Table 2 also contains a calculation of the quantity of RDS if 10 lbs. of each of these corn sources are fed. These large differences in RDS need to be accounted for in formulating rations to maintain an optimum rumen environment and lower the risk of acidosis. A recent review with more details on starch degradability of corn grain is available (Firkins, 2006). High moisture corn was reported to have a

higher RDS than dry corn (Knowlton et. al, 1998). The average RDS of dry ground or rolled corn was 65% versus 84% for high moisture ground or rolled corn. Total tract starch digestibility was also higher for cows fed high moisture corn.

Table 2. Corn grain particle size and ruminal starch degradability ^a

Corn Type	Mean particle size, microns	Effective Rumen Degradability, %	Rumen Degradable Starch, lbs ^b
Cracked	4308	44.6	3.25
Chick cracked	2577	53.3	3.89
Fine ground	686	64.5	4.7
Steam-flaked	2896	75.4	5.5

^a Lykos and Varga (1995)

^b Assumes that 10 lbs. of corn grain containing 73% starch is fed

CORN SILAGE

In many U.S. dairy rations, corn and small grain silages can be a major contributor of ration starch. The average starch content of corn silage samples analyzed by the Dairy One Forage Laboratory between May, 2006 and April, 2007 was 31.9% with a standard deviation of 7.2. This would indicate that about 67% of the samples analyzed had starch levels between 24.7 and 39.1%. A large number of factors influence both the starch content and starch digestibility in corn silage (Weiss and Firkins, 2007). These include hybrid, kernel texture, maturity, processing and storage factors. Starch digestibility generally decreases as maturity and dry matter increase. Kernel processing tends to increase starch digestibility. The benefit of kernel processing is greater with higher dry matter more mature corn silage.

Kernel texture or hardness is an additional factor that affects starch digestibility. In situ ruminal starch degradability was reported as 46.2% for flint corn grain and 61.9% for dent corn grain (Phillippeau et. al., 1999). The vitreousness of the dent corn was 51.4% while it was 71.8% for the flint corn. A higher vitreousness value indicates a harder kernel texture and a lower ruminal starch degradability. In this study, ruminal starch degradability could be accurately predicted from vitreousness ($R^2 = 0.89$). A slightly better prediction of ruminal starch degradability could be obtained by combining apparent density with the 1,000 grain weight. This approach had an R^2 of 0.91. A second study examined the relationship between vitreousness and ruminal in situ starch degradability (Correa et.al., 2002). Dent hybrids from the U.S. and flint hybrids from Brazil were harvested at varying stages of maturity. The vitreousness of the flint hybrids averaged 73.1% while the value for the dent hybrids was 48.2%. The correlation between vitreousness and ruminal starch degradability was – 0.93. A correlation of – 0.87 was reported in this paper between kernel density and ruminal starch degradability.

A number of approaches are being used to “estimate” starch digestibility in corn silage. One method is the use of in situ starch disappearance (Johnston and Tricarico,

2006). The percent of the total starch in the residue after a 6 hour incubation for 350 corn silage samples was 20.4 to 50.2%. Degree of starch access (DSA) is an enzymatic procedure that can be used to determine starch degradation potential (Blasel et.al., 2006). The range in DSA for 22 corn silage samples was 552 to 1043 g/kg of starch with a mean of 834. The challenge is that the results of these analyses have not been calibrated with in vivo animal trials. However, they do have the potential to assist in ranking samples on potential starch digestibility.

A report on the relationship between corn silage storage time and ruminal starch degradability recently became available (Newbold et.al., 2006). This study used 15 corn silages that varied in dry matter and starch content stored in bunker silos on commercial dairy herds. The average dry matter was 33.2% (range was 17.1 to 47.6%). Average starch content was 31.2% with a range of 12.6 to 42.6. Each silo contained one hybrid and was sampled every 2 months from 2 to 10 months after being filled. The percent of the starch degraded after a 3-hour incubation increased from 53.2% at 2 months of storage to 69% after 10 months of storage. There was little increase in starch degradability until 6 months of storage in this study. The quantity of starch from corn silage degraded in the rumen would increase from 2.48 to 3.23 lbs. with a corn silage feeding rate of 15 lbs. of dry matter per day. This additional quantity of ruminal degradable starch could increase the risk of altered rumen fermentation if no adjustments were made in the other NFC and fiber sources in the ration. This study also confirms field observations that corn silage “feeds better” after a few months in storage. It is not known how different hybrids would change in starch degradability with varying storage times.

NONFORAGE FIBER SOURCES

A number of nonfiber forage sources (NFFS) are available that can be used as partial replacements for corn. The most common ones used are soybean hulls, beet pulp, citrus pulp, corn gluten feed and wheat byproducts. Even though these feeds contain low starch levels, they contain high levels of other carbohydrates including sugars, pectin or soluble fiber.

A trial was conducted at Cornell to examine the use of soybean hulls as a partial replacement for high moisture shelled corn in rations for early lactation dairy cows (Stone, 1996). Cows were assigned to treatments at 6 to 10 days post calving. The experimental diets were fed for 98 days. Table 3 contains the results of this trial. There were no significant differences in milk production or milk composition related to ration for the cows in this trial. The multiparous cows consumed more dry matter on the soybean hull ration than cows on the control ration.

A trial at the University of Illinois used pelleted soyhulls at levels of 0, 10, 20, 30 or 40% of the total ration dry matter (Ipharraguerre et.al., 2002). The added soyhulls replaced ground shelled corn in the ration. Ration nonstructural carbohydrate levels decreased from 35.9 to 15.6% as soyhulls were added to the ration. There was a linear

decrease in DMI as the level of soyhulls in the ration increased. However, the majority of this decrease was observed at the 30 and 40% level of added soyhulls. There were no differences in 3.5% fat-corrected milk and milk crude protein yield or percent. Milk fat percent increased as soyhulls were added but milk fat yield was not altered.

Table 3. Soybean hulls in dairy cattle rations ^a

Item	Control Ration	Soybean Hull Ration
Ration ingredients, % of DM		
Corn silage	26.3	26.3
Alfalfa silage	26.3	26.3
HMSC	22.8	8.8
WCS	9.9	9.9
SBM	2.1	1.2
SoyPlus	10.5	11.5
Soybean hulls	0	14.2
Ration composition, % of DM		
CP	18.9	18.2
NDF	29.2	36.3
Starch	25	16 ^b
Animal performance – Multiparous cows		
DMI, lbs/day	45.54	49.72
Milk, lbs/day	89.5	91.7
3.5% FCM, lbs/day	90.6	92.2
Milk fat, %	3.58	3.56
Milk protein, %	2.92	2.89
Animal performance – primiparous cows		
DMI, lbs/day	36.5	37.6
Milk, lbs/day	69.3	69.3
3.5% FCM, lbs/day	71.6	71.7
Milk fat, %	3.7	3.7
Milk protein, %	2.78	2.93

^a Stone, 1996

^b Calculated using the CPM-Dairy program

A number of papers have examined beet pulp or citrus pulp as partial substitutes for corn grain in dairy rations. Voelker and Allen (2003a) used pelleted beet pulp to replace high moisture corn (HMC). The levels of beet pulp in the rations were 0, 6, 12 and 24% of ration dry matter. The level of HMC in these rations decreased from 35.6 to 11.4% of the ration as the beet pulp was added. Ration starch levels decreased from 34.6 (0%

beet pulp) to 18.4% (24% beet pulp). Dry matter intake decreased as the level of beet pulp in the ration increased. However, intakes were similar for cows fed 9, 6 or 12% beet pulp. The cows fed the ration with 24% beet pulp consumed about 4 lbs. less dry matter per day. In this trial, milk production, milk fat and milk protein did not differ between rations. Ruminal starch digestibility decreased as beet pulp was added to the ration (Voelker and Allen, 2003b). Total tract starch digestion was not different between these rations. Total tract NDF digestion increased as beet pulp level in the ration increased. At the levels of beet pulp added in this trial, the efficiency of ruminal microbial protein production was not affected (Voelker and Allen, 2003c).

Rations containing 23.6% citrus pulp (CP) or 25.3% hominy (H) were compared for dairy cattle (Leiva et. al., 2000). The starch levels in these rations were 15 (CP) and 26.5% (H). Dry matter intake and milk production were similar for these 2 rations. A second trial by the same authors used rations containing 20.5% citrus pulp (CP) or 19.5% corn meal (CM). These rations were fed to cows in a group situation. In this trial, cows on the CM ration produced more milk and more pounds of milk fat and protein than cows fed the CP ration. However, cows on the CP ration had a higher milk fat test.

A trial was conducted at Cornell to examine citrus pulp as a partial replacement for corn grain in dairy rations (Solomon et. al., 2000). Rations contained either 9.7 (HS) or 23.9 (HP) % citrus pulp. Corn grain was 23.7 (HS) or 9.3 (HP) percent of the total ration. Ration starch levels were 28 (HS) and 17(HP) %. There were no differences in milk production, milk fat % or milk fat yield between rations. Milk protein % and milk protein yield were higher for the HS ration. A digestibility trial was also conducted with these rations (Miron et. al., 2002). NDF digestibility was higher for the HP ration. Cows fed the HP ration also had higher digestibility of the total neutral detergent soluble carbohydrates and total carbohydrate digestibility.

SUMMARY

1. Neither rumen microorganisms or dairy cows have an actual "starch" requirement.
2. Rumen microorganisms do need rumen fermentable carbohydrates as an energy source and to support microbial protein production.
3. Corn grain contains the highest starch content of feeds typically used in U.S. dairy rations. It may still be an economical feed ingredient for use in dairy rations when considering the nutrients it contributes to the ration.
4. Data from high producing dairy herds indicate that ration starch levels of 17 to 30% starch can support high levels of milk production.
5. Research data indicates that rations containing as low as 15% starch are adequate to support milk production when nonfiber forage sources are used as partial replacements for corn grain.
6. Rumen degradable starch needs to be considered in formulating dairy rations. This is calculated from total starch content and the degradability of starch in the rumen.

7. Ruminal starch degradability of corn grain can vary significantly. Factors such as kernel hardness, particle size, processing methods and storage as high moisture corn can all affect degradability of starch in the rumen. This needs to be considered in putting dairy rations together.
8. The ruminal starch degradability of the starch in corn silage appears to increase with longer fermentation time in the silo. A similar pattern should exist for high moisture corn. This increase in rumen degradable starch may indicate adjustments in the other NFC and fiber components of the ration to lower the risk of acidosis.

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